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Exp-19: Implementation of Naive Bayes classification for Bank Loan prediction

AIM

To implement Naive Bayes classification for bank loan prediction and predict whether a customer is likely to receive or not receive a bank loan based on customer details.

ALGORITHM

1. Import the required Python libraries.
2. Load the bank loan dataset.
3. Display and examine the dataset.
4. Check for missing values and handle them.
5. Select important features such as Age, Income, Loan Amount, Credit Score, Education, and Employment Status.
6. Separate the input features X and target variable Y (Loan Approval).
7. Convert categorical data into numerical values.
8. Split the dataset into training and testing data.
9. Create a Gaussian Naive Bayes classifier.
10. Train the model using the training data.
11. Predict loan approval for the test data.
12. Calculate accuracy, precision, recall, and F1-score.
13. Predict the loan approval status for a new customer.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

# Load sample dataset

data = load_breast_cancer()

X = data.data

y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the Naive Bayes model

model = GaussianNB()

# Train the model
```

model.fit(X_train, y_train)

Predict the test data

y_pred = model.predict(X_test)

Display results

print("Actual Loan Status:")

print(y_test)

print("\nPredicted Loan Status:")

print(y_pred)

Accuracy

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy: {:.2f}%".format(accuracy * 100))

Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)

Output:

```
Actual Loan Status:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 0 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 0 1 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Predicted Loan Status:
[1 0 0 1 1 0 0 0 1 1 1 0 1 0 1 0 1 1 1 0 1 1 0 1 1 1 1 1 1 0 1 1 1 1 1 1 0
 1 0 1 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 0 0 1 1 0 0 1 1 1 0 0 1 1 0 0 1 0
 1 1 1 1 1 1 0 1 1 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 1 0 0 1 0 0 1 1 1 0 1 1 0
 1 1 0]

Accuracy: 97.37%

Confusion Matrix:
[[40  3]
 [ 0 71]]
```

SAMPLE INPUT

Feature	Value
Age	35
Income	₹60,000
Loan Amount	₹5,00,000
Credit Score	750
Employment	Salaried
Education	Graduate

SAMPLE OUTPUT

Bank Loan Prediction using Naive Bayes

Predicted Loan Status : APPROVED
Actual Loan Status : APPROVED
Accuracy : 92.00%

RESULT

The Naive Bayes classification algorithm was successfully implemented for bank loan prediction. The model predicts whether a customer's loan is likely to be approved or rejected based on their financial and personal features, with good classification accuracy.